**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# LESSON 221 · A*: heuristic reliability and route evaluation

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. Calculate f when g=4 and h=3.

Q2. What is the role of a predecessor table?

Q3. If S–A=2 and A–G=5, what is this route's total cost?

Q4. The true cheapest remaining cost from A is 4. Is h(A)=6 admissible?

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# I Do · Read new knowledge

## Key vocabulary

admissible — Heuristic never overestimates the true cheapest remaining cost.

consistent — Heuristic obeys h(u) ≤ edge(u,v)+h(v) for every edge.

suboptimal — A valid route whose cost is greater than the cheapest possible.

frontier — Discovered vertices awaiting expansion.

heuristic — An estimated remaining cost used to guide search.

## New learning

Find the real remaining cost — From A, the cheapest route A→B→G costs 2+2=4. Therefore h(A)=3 is admissible, but h(A)=9 is not.

Safe estimates support optimal search — With h(A)=3, A is expanded before B, improving the route to B and finally G at cost 6.

Watch the misleading estimate — With h(A)=9, A has f=11; B has f=7 and leads to goal at cost 8.

Why the error matters — A* that stops on the first goal removed from the open set can return this 8-cost path before exploring A, even though a 6-cost route exists.

Check the guarantee — Admissibility means no overestimate. Consistency adds the edge test h(u) ≤ weight(u,v) + h(v). Without reopening settled nodes, A* requires consistency for its shortest-path guarantee. Explain this as well as the route cost.

## Worked example

Check admissibility and trace U→G with safe estimates.

1. Predict the choice — V has g1,h3,f4; W has g5,h1,f6. Expand V.

2. Review the choice — V improves W to g3 and f4; expand W.

3. Review the choice — W improves the goal G to g5,h0,f5.

4. Review the choice — The path is U→V→W→G with total cost 5; the heuristics do not exceed the actual remaining costs.

# We Do · Practise with guidance

Can a bad estimate make A* return a longer route?

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Edges M–N=2, M–O=5, N–O=2, N–G=7, O–G=2.
h(M)=4,h(N)=7 (unsafe),h(O)=1,h(G)=0.
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Predict the choice

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

After M, O has f5+1=6; N has f2+7=9, so O is selected.

Review the choice

O reaches G at g7,f7. G is selected before N, so the early-goal algorithm returns cost 7.

But M→N→O→G costs 2+2+2=6; the bad h(N)=7 overestimates N's true remaining cost 4.

Using an admissible estimate such as h(N)=3 allows the cheaper path to be found.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Question 1

The true remaining distance from node A to goal G is 6. Decide whether h(A)=4 and h(A)=8 are admissible, with reasons.

### Question 2

Edges S–A=2, S–B=6, A–B=2, B–G=2, A–G=8. Calculate the best S→G route and explain why h(A)=9 can mislead A*.

### Question 3

Compare Dijkstra's algorithm and A* in their selection rule, heuristic use and guarantee of shortest paths.

# Exit Ticket · Answer before checking

If a heuristic overestimates the true remaining cost, A* may lose its shortest-route guarantee.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. f=7.

Q2. It lets us reconstruct the path by following predecessors backwards.

Q3. 7.

Q4. No. It overestimates the true cost.

### You Do pages · Independent questions

Q1. [4] h(A)=4 is admissible [1] because it does not exceed 6 [1]; h(A)=8 is inadmissible [1] because it overestimates 6 [1].

Q2. [6] Route S→A→B→G [2], cost 6 [1]; from A the true cheapest remaining cost is 4 [1], so h(A)=9 overestimates [1] and can delay expanding A while a more expensive goal route appears [1].

Q3. [6] Dijkstra selects lowest tentative g [1] and does not need h [1]; A* selects lowest f=g+h [2]; Dijkstra needs non-negative edges [1]; A* requires suitable admissible heuristic (and appropriate closed-set handling, usually consistency) to guarantee optimality [1].

### Exit Ticket · Model answer

True. An inadmissible estimate can make a less costly path appear too unattractive to explore before stopping.

# Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

| Do Now | I Do | We Do | You Do | Review & Improve |
|---|---|---|---|---|
| Recall previous learning | Read new knowledge | Practise with guidance | Work independently | Check understanding |